

The Language of Business

Analytics

Foundational Concepts of Analytics

01) Business Analytics:

- Business analytics is the process of using data and technology by investigating business trends, underlying causes for decision making.

Example - Shoprite bread is sold more on a Sunday, Therefore the business analytic will suggest to put the Albany bread in front of the store and add butter/margarine promo next to it, this will increase profit every Sunday.

IBM (2023) What is business analytics? Available at: <https://www.ibm.com/think/topics/business-analytics> (Accessed: 25 September 2026)

02) Data Analytics:

- The task in collecting, cleaning, analysing data to find trends and patterns, that help with decisions, and report on data that is analysed.

Example - Data analyst will collect data from the Shoprite data warehouse, The data analyst will analyse which products that were sold the most and which products (coffee) was not sold. They will then suggest ways to make the coffee sell by decreasing price, push for a sale. The data analytics tell you what happened.

03) Business Intelligence (BI)

- Is the process and technology of gathering past and present business data to turn it into easy/understandable reports, like charts, diagrams.

Example - The CEO will ask to see progress on sales between, bread, milk and rusks, Business Intelligence analyst will bring a bar graph or Pie chart to make it easy for the CEO.

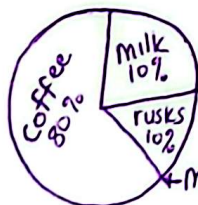
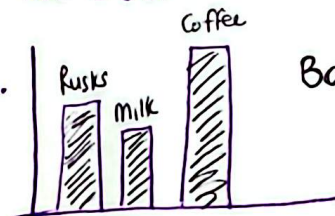


Diagram of Products That are Sold in the month of September.



Bar graph

IBM (2023) What is business intelligence? Available at:

<https://www.ibm.com/think/topics/business-intelligence> (Accessed 18 September 2026)

Q4) Data - Driven Decision Making:

- Making decisions using data, and not making decisions with your gut feeling. Making organised decisions based on data analysis and facts.
- Example - Member in the community posts that there will be a water outage at 18:00pm and posts the message in the group at 07:00am. Because last time she posted a warning message at 11:00am only 5 viewed out of 50 people, this is during working ~~hours~~ hours. However Sharon saw that people view messages in the morning with 49 people viewings at 07:00 rather than 11:00am.

IBM (2024) What is data-driven decision-making?
Available at: <https://www.ibm.com/think/topics/data-driven-decision-making> (Accessed: 20 September 2026)

Q5) Big Data:

- Big data is too big, very fast and messy. This includes structured, semi-structured and unstructured data.

Example - Small data is my contact list on my phone with 150 numbers

Big data - When Vodafone or Telkom's data with calls, SMS, air-time data usage for 40 million customers everyday.

Google Cloud (nodate) What is big data? Available at:
<https://cloud.google.com/learn/what-is-big-data> (Accessed 21 September 2026)

Q6) Descriptive Analytics:

- This is a process of taking historical data by understanding what has happened in the past, using reports, charts and totals

Example - A Content Creator checking number of views on social media past week.

Lindi

Munday - 1000 views
Tuesday - 3000 views
Wednesday - 200 views
Thursday - 900 views
Friday - 5000 views

Lindi then does a descriptive analysis that The

Total views this week were = 10 100

Average view per day = 1 800

Highest view = Friday - 5000 views

Lowest views = Wednesday - 200 views

Friday was the 500% higher than other days.

Harvard Business School online (nodate) Descriptive Analytics. Available at:

<https://online.hbs.edu/blog/post/descriptive-analytics> (Accessed 21 September 2026)

Q7) Diagnostic Analytics:

Why it happened, It is the process where you analyse historical data to understand the cause or reason why it happened.

Example - Lindi has rather identifies that she has 5000 views on Friday why? because she posted at 12:00 pm during lunch time, most Fridays it's half a day for school pupils, employees, so she will get more views than the other days.

IBM (2025) Diagnostic Analytics? Available at:

<https://www.ibm.com/think/topics/diagnostic-analytics> (Accessed 21 September 2026)

08) Predictive Analytics:

- The use of historical and current data, statistical modeling to predict future results.

Example: If Lindi the content creator post her videos every Friday at 12h00pm with creative, fun content, she predicts she will get 5000 views again.

Or more due to her past analysis.

IBM (2023) Predictive Analytics? Available at: <https://www.ibm.com/think/topics/predictive-analytics> (Accessed: 21 September 2026)

09) Prescriptive Analytics:

- What actions should be taken, by using data and models to recommend the best most effective course of action to achieve the outcome.

Example - Lindi's action from her analysis that every Friday at 12h00pm

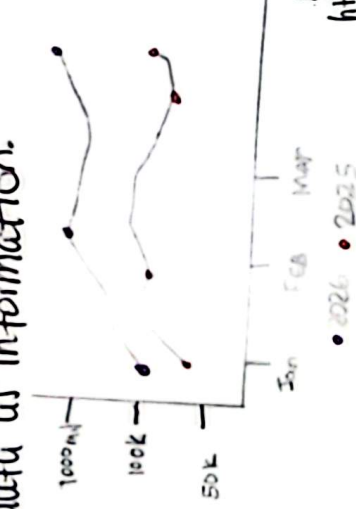
She will upload her videos with content, more engagement to get

more views and ~~content~~ posting on Monday - Thursday.

IBM (2023) What is Prescriptive Analytics? Accessed at: <https://www.ibm.com/think/topics/prescriptive-analytics> (Accessed: 22 September 2026)

10) Data Literacy:

- The ability to read, understand/comprehend by utilising data can read a chart, or a bar graph. Being able to communicate data as information.



2025-2026 Profit made by BNG wine

- In 2025 BNG made less than 100 thousand however in 2026 made more than 1 million in sales the graph shows the fluctuation of profit made throughout the year of 2025 and 2026.

IBM (2025) Data literacy Available at:

<https://www.ibm.com/think/topics/data-literacy> (Accessed 22 September 2026)

DATA, MEASUREMENT & REPORTING

11) Structured Data:

- Is information organised into neat rows and columns with fixed formats. It is high quantity, easy to search.

Example - Excel spreadsheets, Bank accounts, Brightlearn (no date) Data Analytics - Course notes. Available at: <https://portal.brightlearn.co.za/module/data-analytics> (Accessed 22 September 2026)

12) Unstructured Data:

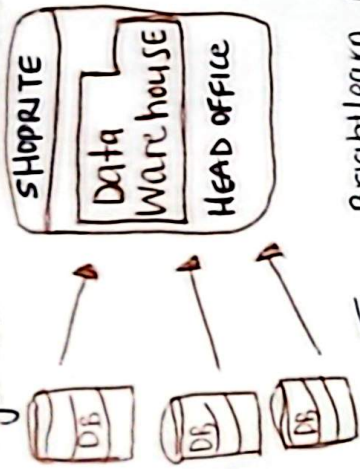
- Is information that lacks a predefined data model or table structure

Example - Emails, social media posts, videos

Brightlearn (no date) unstructured data data Analytics - Course notes. Available at: <https://portal.brightlearn.co.za/module/data-analytics> - September - 3030/content/notes (22 September 2026)

13) Data Warehouse:

- A data warehouse is a central system that stores, cleans, and organizes historical data that's pulled from many source systems.



BrightLearn (no date) Data Analytics - Course Notes, Available at:

[<https://drive.google.com/file/d/1R27UOIUR6K9U0hhTHIVoumSncHIF4v2/view>] (Accessed 22 September 2026)

14) Data Mining:

- Data Mining is finding hidden patterns in a number of data that one individual would not see by looking at it. It is like digging through tons of sand to find small pieces of diamonds.

Example - Department of education has 200,000 comments on Instagram and communications team can not go through and read all comments, data mining will find hidden patterns that 70% of people complain about learners not receiving textbooks on time or principals not receiving money for norms and standards that live in KZN and complain at ashooam - 10hooam.

Communications did not see ~~where~~ which province or time people are complaining, with data mining it picked up the province and times.

[Available at: <https://www.snowflake.com/en/fundamentals/cloudmining/>] (Accessed 22 September 2026)

Province and times. Snowflake (no date) What is data mining?

15) Data Visualization:

- Is presenting data through pictures and not data, such as graphs, pie charts, bar graphs. This is to make it easy to understand data and find patterns.

Example - Pie chart on sales that Foschini group made in the month of August 2026. The pie chart shows parts (50% sales in bags, 30% sales in clothing, 20% sales in shoes).



This is to make it easy for the management to understand.

IBM (no date) Data Visualization? Available at:

[<https://www.ibm.com/think/topics/data-visualization>] (Accessed 22 September 2026)

16) Dashboard:

- Dashboard is a visual tool that displays key metrics and KPIs on one screen to check performance.

Example - You buy a car. You do not ~~but~~ open the bonnet to check the engine to look at speed, fuel, and heat. The car has a screen in front of you that displays kilometers, speedometer, fuel, when you indicate. The dashboard is the screen displays tools.

17) Metric :

- It is numbers you observe, used to track and assess performance.

Example - In the office ^{umgeni water} people wore Red, Pink or black and red for Valentines day. The METRIC is 20 wore Red, 10 wore Pink and 11 wore black and red. The metric is the number that's been observed.

IBM (2021) Metrics - IBM documentation Available at: <https://www.ibm.com/docs/en/iis/11.3.0?topic=Methodology-metrics> (Accessed 24 September 2026)

18) Key Performing/Performance Indicator (KPI) :

- The important integers that tell you if you are winning or losing an important measurement value that show how effective an organisation is achieving or not.

Example - Umgeni water ~~is~~ to make sure that its customers are happy and have water. They have a Instagram page to post the latest news updates etc. The key Performance Indicator is how well they respond on time to complaints and number of complaints they resolve. If the community has water. If the audience responds well to posts and less complaints. Then umgeni water has achieved.

Investopedia (2005) KPIs : Key Performance Indicator ? Available at: <https://www.investopedia.com/terms/k/kpi.asp> (Accessed 22 September 2026)

19) Bench Mark :

- The line you compare yourself against. Your performance can be measured and compared by the standard you set.

Example - Thandi is a grade 11 student. The benchmark to pass is 90% in exams. Thandi in her last exam in first term she got 70% which is lower than the set bench mark. Thandi will have to study more to achieve the 90% benchmark.

IBM (2017) Define benchmarks : Available at: <https://www.ibm.com/docs/en/iis/11.7.0?topic=techniques-defining-benchmarks> (Accessed 23 September 2020)

20) Segmentation :

- Cutting one big group into small groups, It is dividing large market or audience into smaller distinct groups each have small characteristics to target them.

Example - Department of education has 20 000 followers, big group of people. Some are angry parents, some are happy learners or ^{teachers} ~~audience~~, others want to enquire. The angry parents are based in Durban, the happy learners and teachers are in Newcastle and those that are enquiring are in Pietermaritzburg. You need to group them because you can not respond the same to all of them.

Location : Durban, Newcastle, Pietermaritzburg

character : Happy, angry and, Those that enquire

AGE : (18-24), (26-50)

} This will help in responding to everyone because you have grouped each audience.

CUSTOMER & SALES KPIs

21) Conversion Rate :

- It is the Percentage of users who take a desired action out of the total number of users.

Example - Cotton On sends a link to get 20% off on items in ~~the~~ email addresses of only 2000 people. See the link and 80 people open the link.

$$80 \div 2000 \times 100 = 4\%$$

only 4% people converted who opened

Shopify (2026) Conversion Rate: Benchmark the link.
bpx (2026). Available at: <https://www.shopify.com/blog/e-commerce-conversion-rate> (Accessed: 23 September 2026)

22) Customer Acquisition Cost (CAC) :

- Amount of money spent in gaining a new customer. The total cost of acquiring a new customer, marketing and sales expenses. If your (CAC) is higher than what your customers pay you make a loss.

Example - I sell sweets in a packet. I spend R40 on printing the posters

and buy data to engage with my customers. I get 10 new clients

who want to buy my sweets. $R40 \div 10 = R4$ per customer in making

profit.

IBM (2024) Customer Acquisition Cost (CAC) Available at: <https://www.ibm.com/think/topics/customers-acquisition> (Accessed 23 September 2026)

23) Customer Lifetime Value (CLV/LTV) :

- How much a customer spends on your product, for their whole life with you. The revenue a business expects to earn from a customer throughout their whole relationship.

Example - (CAC) is the cost to get a customer, (CLV) - is the value you get from a customer.

If a customer buys sweets from me 2 times a week = $R12 \times 2 = R24$
for a whole year ~~52 weeks~~ 40 weeks $R24 \times 40 \text{ weeks} = R960$ a week

The student will give you R960 for the year.

IBM (2024) Customer Lifetime Value (CLV) Available at: <https://www.ibm.com/think/topics/customer-lifetime-value>
Accessed (23 September 2026)

24) Churn Rate :

- Number of customers that leave, number of customers that stop purchasing a product over a period of time.

Example - I had 20 customers and lost 3 due to Judy selling candy floss

$$3 \div 20 \times 100 = 15\% \text{ Churn}$$

Investopedia (2004) Churn Rate. Available at: <https://www.investopedia.com/terms/c/cr>
ChurnRate.asp (Accessed 23 September 2026)

25) Customer Retention Rate:

- How many customers stayed. Number/percentage of customers a business retained (kept) over a specific time period.

Example - I have 20 customers + 10 new customers

$$20 - 10 = 10 \text{ of original}$$

$$10 \div 20 \times 100 = 50\% - \text{Retention}$$

IBM (2024) Customer retention rate? Available at: <https://www.ibm.com/think/topics/customer-retention-rate> (Accessed 23 September 2024)

26) Net Promoter Score (NPS):

- The number of people that will promote you to friends, family or colleagues. Customer that keep their loyalty and are satisfied with your product that they recommend.

Examples: Ask my fellow classmates if they would recommend my sweet packets I asked 50 students, 25 students give a out of 10 they will promote

$$20 \text{ are Positive and } 5 - \text{will not. } 25 - 20 = 5$$

IBM (2024) Net Promoter Score? Available at: <https://www.ibm.com/think/topics/net-promoter-score> (Accessed 23 September 2024)

27) Average Order Value (AOV):

- average money a customer spends in one order

Eg I made R 400 from sweet packets I had 20 orders

$$AOV = R 400 \div 20 = R 20 \text{ Student spends per day}$$

Shopify (2025) Average order value: Formula and 7 ways (2025), Available at: <https://www.shopify.com/blog/average-order-value> (Accessed 23 September 2025)

28) Sales Funnel:

IBM (2025) Sales Funnel? Available at: <https://www.ibm.com/think/topics/sales-funnel> (Accessed 23 September 2025)

- The process from stranger to buyer. This is a visual representation of a customer's journey from first contact to making a purchase.

Example - My sweet packets exist at res and on campus. So see my posters, some become interested and ask about my sweet packet, with the information they receive about my sweet packet and what is inside, they will purchase, 50 see poster. 10 stop and ask, 25 ask and 15 purchase.

29

Marketing & Campaign Analytics

29) Marketing Campaign:

- Planned action to get customers. It's a coordinated set of activities designed to promote a product, service or brand to achieve a goal in business a campaign has:

- Goal - Sell Renda sweet packets 300 sweet packets and gain 100 new customers
- message - Best sweet packet with different kind of sweets
- channel - Posters, status update on whatsapp, campus radio
- Time - For 3 weeks
- Budget - R 200 packages, R 40 posters R 20 airtime

Big Commerce (nodate) Marketing Campaign? Available at: <https://bigcommerce.com/glossary/marketing-campaigns/>

34) Click-Through Rate (CTR) %

- A percentage of people who click on a ad or link after seeing your advertisement or link. To measure the effectiveness of the ad.

Example - My post on facebook about my sweet packets.

1200 viewed my ad on facebook

50 clicked on order now

CTR = $50 \div 1200 \times 100 = 4.1 = 5\%$ (rounded off)

WordStream (2024) <https://www.wordstream.com/click-through-rate> (Accessed: 24 September 2026)

35) Bounce Rate %

- People who came and left immediately. The percentage of visitors who leave a website after viewing without taking action.

Example - 100 people visited my facebook ad or site where you order 60 people leave after few minutes with no click to order

$60 \div 100 \times 100 = 60\%$ - High bounce rate

Google analytics help (no date) [UVA] Bounce rate [legacy]. Available at:

<https://support.google.com/analytics/answer/1004909> (Accessed 24 September 2026)

36) A/B Testing %

- Testing products to see which one is most likely to be sold more than the other one. More of a experiment two different products same item, to see which one performs better.

Example - You make milkshakes and take a photo and sell on Instagram/facebook

① Cheilatte milkshake with chocolate sprinkles

② Gourmet chocolate cookies & Cream with cookie bits.

1) Send a ~~5~~ picture ① to 5 people and picture ② to another 5 people
Picture ① gets 2 orders and picture ② gets 4 orders

Picture ② The gourmet chocolate cookie & Cream with cookie bits get the most order.

Salesforce (no date) What is A/B Testing? Available at: <https://www.salesforce.com/ap/marketing/what-is-a-b-testing/?bc=OTH> (Accessed 24 September 2026)

37) Lead Generation %

- Attracting customers who will buy later. The customer shows interest in your product however will not buy now but later.

Example - I put a post on Instagram that says "Click on the link and get Menu for your delicious Gourmet milkshakes". This makes you customer check your menu even though they will not buy now.

Search engine land (2026) lead Generation? Definition, Process, and Example. Available at: <https://searchengineland.com/guide/what-is-lead-generation> (Accessed: 24 September 2026)

38) Marketing Attribution %

- Who will get credit for the sale made. Identifying which marketing channels or touch points that contributed to conversion and credit accordingly.

Example - Jenna sees my gourmet milkshakes on, whatsapp, Instagram and facebook! Then Jenna's friend Sarah tells her about the delicious milkshakes she tried. Jenna then buys the gourmet chocolate cookie & Cream.

Adobe (2026) marketing attribution - Available at:

<https://business.adobe.com/au/blog/basics/marketing-attribution> (Accessed: 24 September 2026)